

JACOB COOK

Security Engineer (L2) | Threat Detection & Response | Anomaly Detection · Python · Linux · MITRE ATT&CK · Cloud IR

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Detection-focused security engineer with hands-on experience triaging events at Level-2 depth, authoring Sigma rules mapped to MITRE ATT&CK adversarial TTPs, and building Python automation for incident response workflows. Comfortable with Linux internals, network analysis, and anomaly detection patterns (regex, stack counting, baseline-difference logic) used in Wazuh lab work. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress. Comfortable with on-call rotations and high-pressure triage.

TECHNICAL SKILLS

Detection & IR: L2 Triage, Incident Response Workflows, Forensic Investigation Concepts (host & network), Sigma Rule Authoring, MITRE ATT&CK TTPs, Anomaly Detection (Regex, Stack Counting, Baseline Lookups), Threat Hunting
Scripting & Automation: Python, PowerShell (familiar), Bash, FastAPI, Webhook-Driven Playbooks, Response Automation
Systems & Cloud: Linux (Ubuntu/CentOS), Windows Endpoints, Wireshark, Azure / Azure AD / Entra ID, AWS labs, SSO / RBAC, Terraform
SIEM & Tooling: Wazuh SIEM (build & admin), Microsoft Sentinel (familiar), KQL/SPL (familiar), VirusTotal, OSINT

TECHNICAL PROJECTS

- Stryker / Intune Detection Pack: Detection-as-Code** 2026
- Authored 6 Sigma-format detection rules for adversarial TTPs mapped to MITRE ATT&CK with anomaly logic (baseline lookups, regex matches, stack-count style suppression) that eliminated 100% of false positives in a 100-event simulation.
 - Documented 3 benign scenarios per rule (18 total) so analysts can triage at L2 depth confidently without escalating known-good activity.
 - Verification: 43 passing pytest cases in repo (tests/test_rules.py).
- Blue Team SOC Monitoring Lab: Wazuh SIEM** 2026
- Built a Wazuh SIEM stack from scratch and validated 12 detection rule patterns against 242 simulated auth-style events with 100% precision after tuning — direct hands-on with security sensors, log normalization, and detection efficacy work.
 - Reduced false-positive volume by 100% by tuning out 79 benign-only matches without dropping a single true positive.
 - Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).
- SOAR-lite: Incident Response Orchestrator** 2026
- Built Python/FastAPI automation for IR workflows — webhook ingestion, playbook routing, IP blocking, Slack notification, and incident documentation — reducing triage time by over 60% in lab testing.
 - Eliminated four manual triage steps and produced documented L2-ready handoff format suitable for high-volume cyber defense operations.
 - Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against real FastAPI app, not mocks.
- Phishing Analysis Lab** 2026
- Triageed 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction — 100% accurate verdicts against ground truth.
 - Documented attacker infrastructure mapping and IOCs in the same evidence-preservation format incident investigations require.
- G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)** 2024
- Engineered RBAC and Azure AD SSO controls — direct cloud architecture and identity security work — that enabled executive-level adoption of a production AI platform.
 - Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

- Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response** 2019 - 2025
- Maintained 99.5% system uptime by triaging telemetry alerts on rotating shifts — direct experience operating in a 24x7 on-call coverage model.
 - Resolved 200+ anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30%.
 - Reduced mean time to recovery by an estimated 20% via cross-team incident coordination, escalation, and post-incident reporting.

EDUCATION & CERTIFICATIONS

- WGU | MS, Cybersecurity (In Progress)** Expected Oct 2026 · GPA 3.50
- CSUS | BS, Computer Science** Jan 2025 · GPA 3.50
- CompTIA:** Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)